

Customer Shopping Behavior Analysis

1. Project Overview

This project analyzes Diwali Sales using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

2. Dataset Summary

- Rows: 11251

- Columns: 15

- **Key Features:**

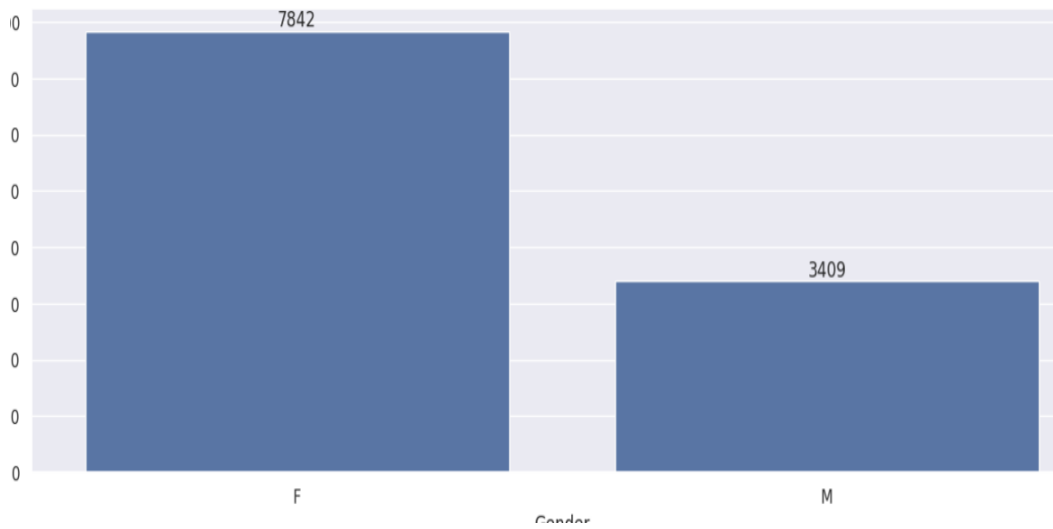
- Customer details -User ID, Cust_name, Gender, Age Group, Age, Marital Status, State, Zone, Occupation, Product Category, Orders, Amount
- Purchase details -Product ID, Product Category, Orders, Amount
- Missing Data: 12 values in Amount column
- Status, Unnamed1 column contains only null value so deleted those columns.

3. Exploratory Data Analysis using Python

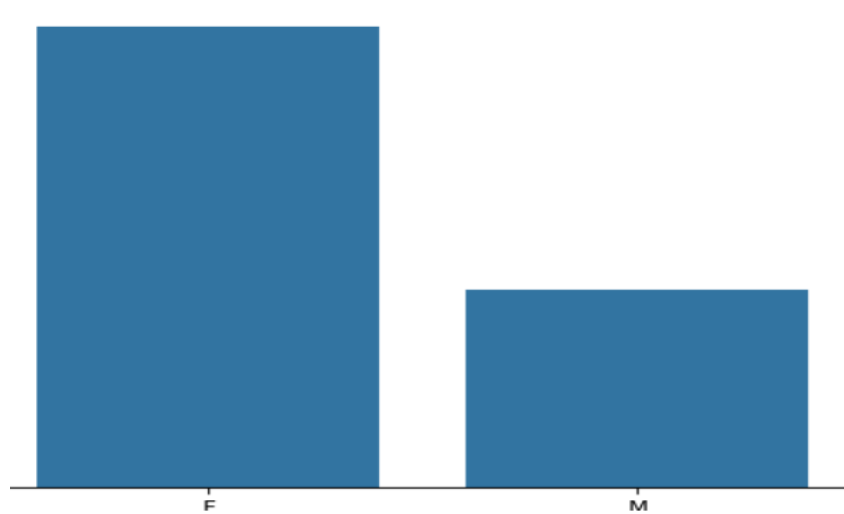
We began with data preparation and cleaning in Python:

- **Data Loading:** Imported the dataset using `pandas`.
- **Initial Exploration:** Used `df.info()` to check structure and `.describe()` for summary statistics
- **Missing Data Handling:** Checked for null values and drop the columns which have all null values
- **Rename:** Renamed Marital status to Shadi.

plotting a bar chart for Gender and it's count-

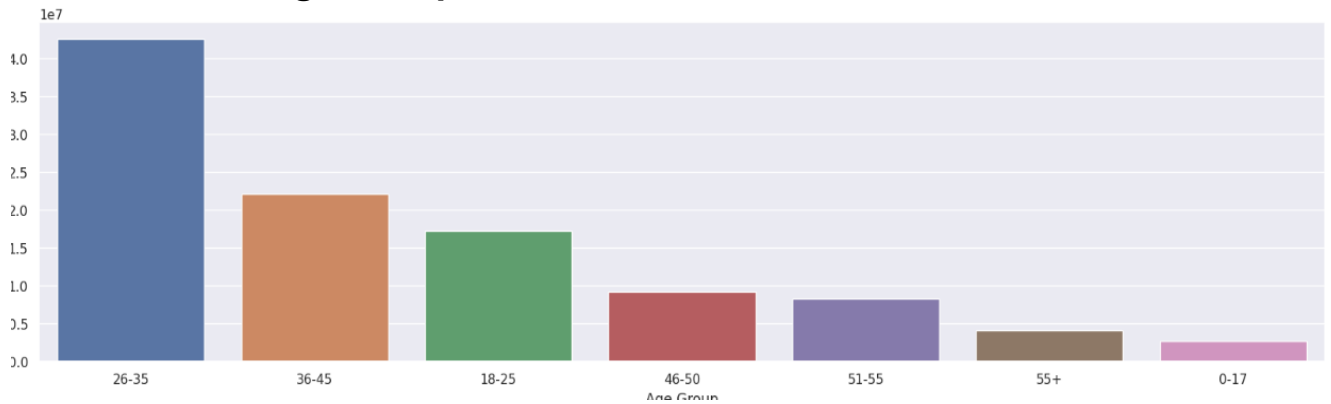


plotting a bar chart for gender vs total amount-



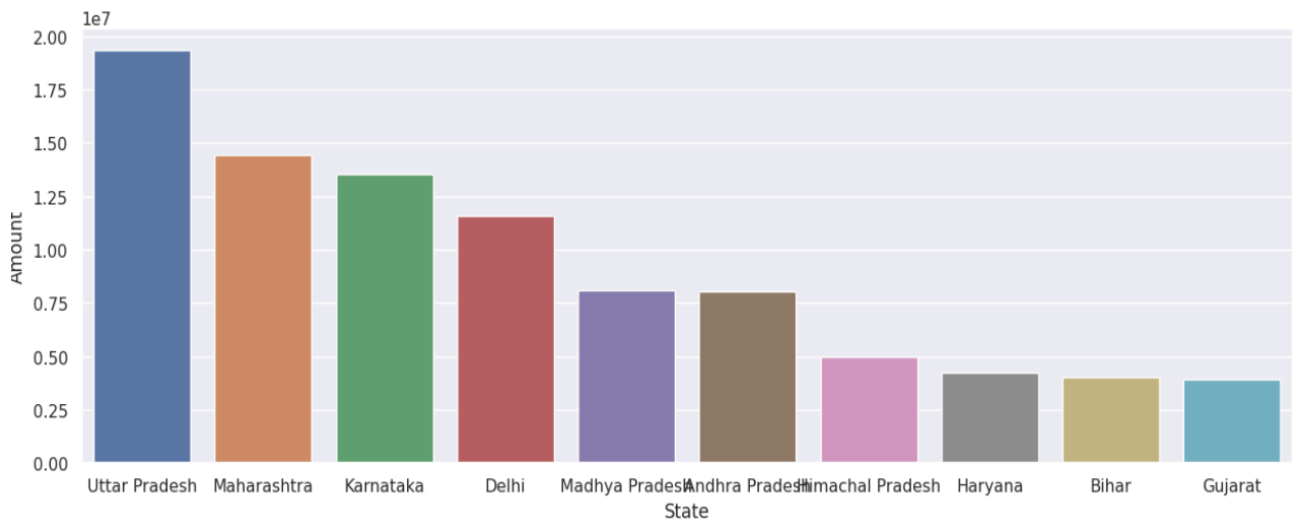
From above graphs we can see that most of the buyers are females and even the purchasing power of females are greater than men

Total Amount vs Age Group



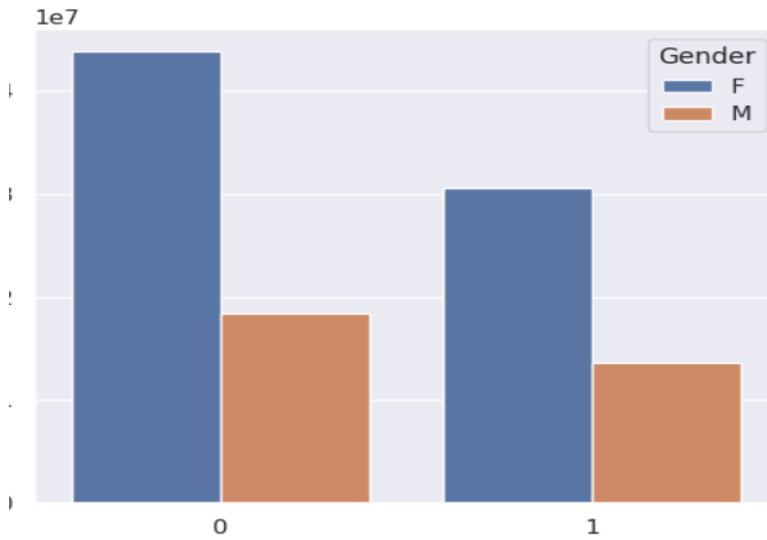
From above graphs we can see that most of the buyers are of age group between 26-35 yrs female

total amount/sales from top 10 states



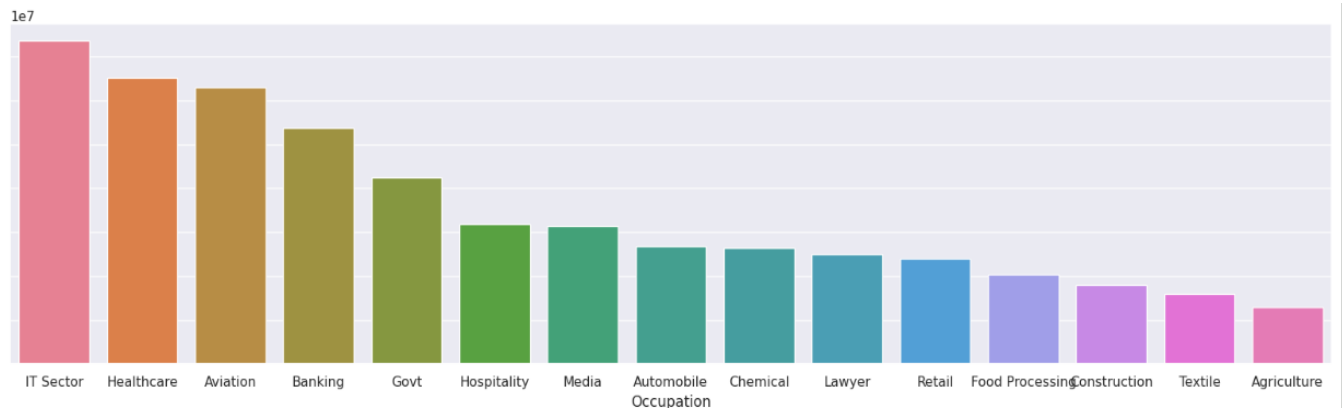
From above graphs we can see that most of the orders & total sales/amount are from Uttar Pradesh, Maharashtra and Karnataka respectively

Marital status and gender vs amount



From above graphs we can see that most of the buyers are married (women) and they have high purchasing power

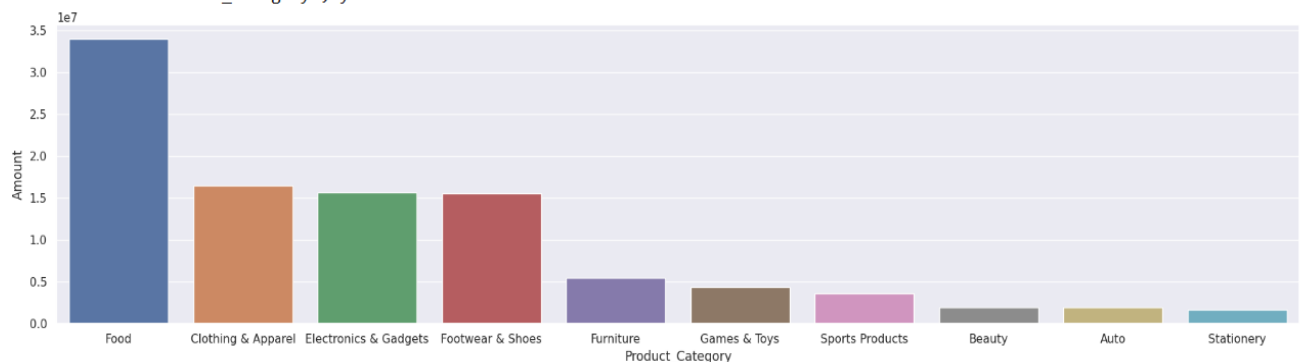
Occupation



From above graphs we can see that most of the buyers are working in IT, Healthcare and Aviation sector

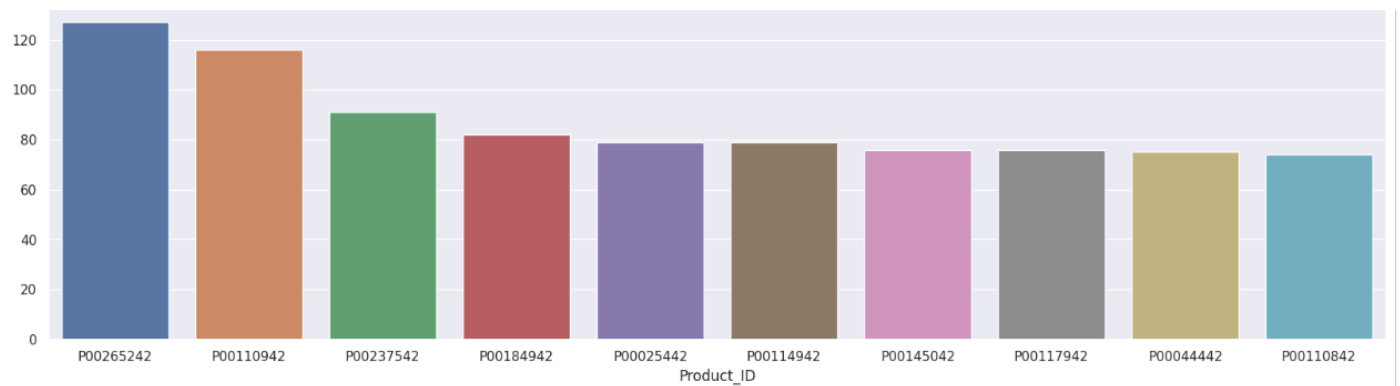
Product Category

Axes: xlabel='Product_Category', ylabel='Amount'>



From above graphs we can see that most of the sold products are from Food, Clothing and Electronics category

Product ID



Conclusion:

Married women age group 26-35 yrs from UP, Maharastra and Karnataka working in IT, Healthcare and Aviation are more likely to buy products from Food, Clothing and Electronics category